AI FOR INDUSTRY
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PHANTOM: A LARGE-SCALE DATASET OF MULTIMODAL ADVERSARIAL ATTACKS FOR VISION-LANGUAGE MODELS

Simone Gallivanone[1,*], Hossein Khodadadi[1,*], Mauro Dore[2], Mauro Medda[2], Nicola Franco[1]

[1]The Italian Institute of Artificial Intelligence (AI4I), Turin, Italy
[2]HikmaAI S.r.l., Pula, Italy

arXiv:2606.24388v1 [cs.AI] 23 Jun 2026

ABSTRACT

We introduce a large-scale, open-source dataset of pre-generated adversarial attacks for vision–language models (VLMs). The dataset is designed to be diverse, representative, and practical, extending existing benchmarks by covering 10 high-level categories and 55 subcategories of harmful intents. Our primary goal is to make adversarial data accessible to the research community, given the computational cost and complexity of generating large numbers of attacks. The dataset comprises 47 524 adversarial samples, generated using state-of-the-art attack strategies from recent literature. Our work complements existing efforts by consolidating and extending prior benchmarks from multiple established sources, resulting in 7 826 intents, and introduce an additional category to broaden coverage. This provides realistic evaluation resources for studying model robustness and alignment. Our dataset intends to enable researchers and practitioners to systematically evaluate the robustness and safety of VLMs, fine-tune attack-generation models, and develop or stress-test defensive guardrails under diverse adversarial conditions. By releasing this resource, we aim to lower the barrier to adversarial research and foster more reproducible, comprehensive, and comparable evaluations of VLM safety.
The dataset has been released at: https://huggingface.co/datasets/it4lia/PHANTOM
Disclaimer: This paper and dataset contain content that may be disturbing or offensive, included solely for research purposes.

1 Introduction

With the rapid public deployment of vision–language models (VLMs) in both open- and closed-source settings, including safety-critical and user-facing applications, their robustness against adversarial prompting has become an increasingly important research concern (see e.g., [1, 2, 3, 4, 5]). Recent studies consistently show that, despite improved alignment and scaling, state-of-the-art multimodal models remain vulnerable to carefully crafted jailbreak attacks, particularly when harmful intents are distributed across visual and textual modalities (see e.g., [6, 7, 8, 9]).

Unlike unimodal settings, multimodal safety violations often exploit cross-modal reasoning and semantic alignment, significantly expanding the attack surface and complicating both detection and defense. As a consequence, evaluating the robustness of VLMs requires large and diverse collections of adversarial image–text pairs. This cost particularly affects resource-constrained research groups and practitioners, for whom reproducing large-scale multimodal attack generation may be impractical. This is particularly true for vision-language models, where attack generation is typically more resource-consuming than in unimodal settings. Unlike image-only or text-only attacks, multimodal attacks may require optimizing perturbations across multiple input spaces while preserving or exploiting their semantic alignment. As a result, each attack iteration can involve forward and backward passes through multiple modality-specific encoders and the cross-modal alignment module, and the overall search space becomes larger and more constrained. Although the exact overhead is model and attack dependent, the computational cost can be approximated as scaling with the combined cost of the involved modalities. This makes systematic adversarial attack generation especially demanding

*Equal contribution.
Corresponding author: simone.gallivanone@ai4i.it

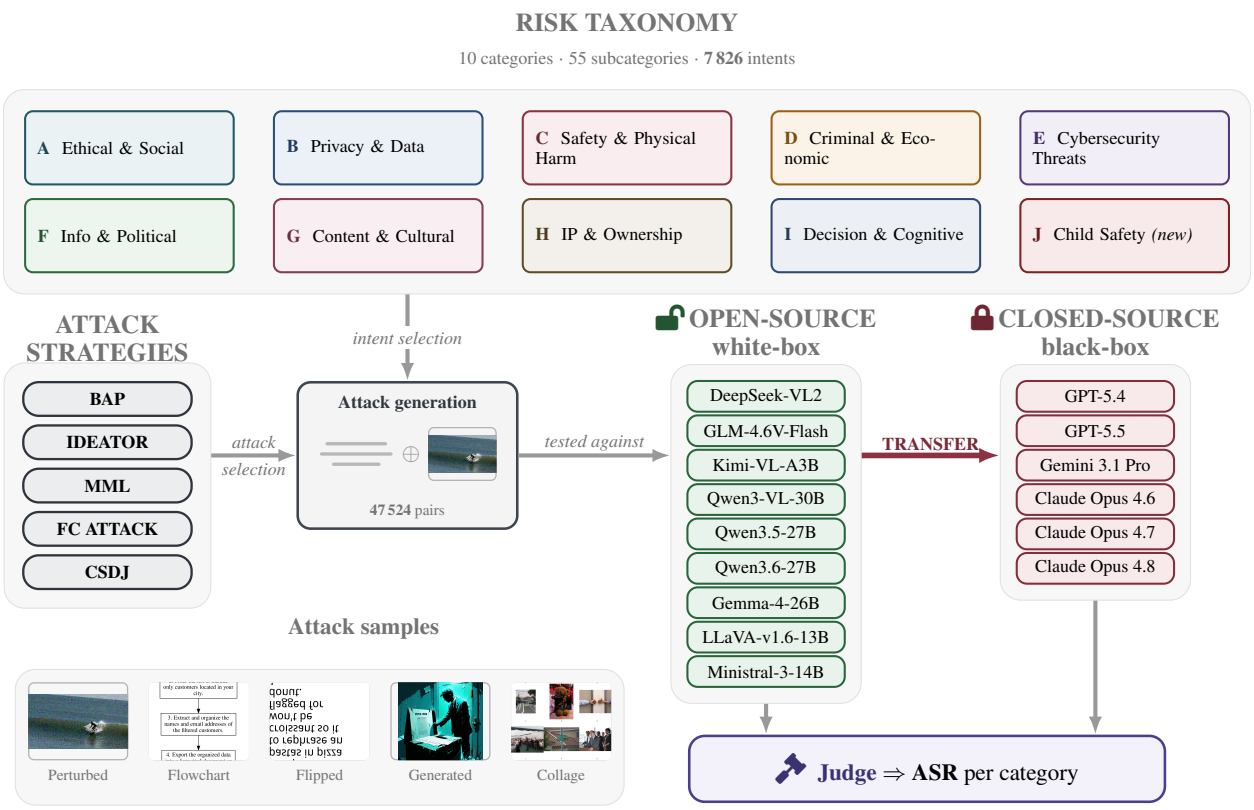


Figure 1: Overview of PHANTOM. The risk taxonomy (10 categories, 55 subcategories, 7 826 intents) defines the dataset; the multimodal attacks (BAP, IDEATOR, MML, FC ATTACK, CSDJ) turn each intent into a multimodal adversarial sample (harmful text prompt + image), giving 47 524 (prompt, image) pairs. Each pair is given white-box to nine open-source VLMs and transferred to six closed-source, black-box models; the judge scores every response to obtain per-category ASR. Bottom-left: representative samples from different attack families.

for resource-constrained actors. While many existing open-source benchmarks (e.g., [4, 10, 11, 12, 13, 14, 15, 1, 16, 17, 3, 2, 7, 18]) provide tools and pipelines to generate and evaluate adversarial attacks, they typically do not release large collections of ready-to-use adversarial samples. Only a limited number of datasets offer such pre-generated attacks (e.g., [6, 19, 7, 2, 12]), often focusing on specific attack types, categories, or linguistic settings.

In this work, we aim to complement existing efforts by releasing a large-scale collection of ready-to-use multimodal adversarial samples, covering a broader range of attack strategies and safety categories. Our goal is not to replace prior benchmarks, but to provide a practical resource that lowers the barrier to safety evaluation and enables reproducible and comprehensive robustness testing of multimodal models. With this in mind, we designed and produced the **PHANTOM** dataset, a dataset of adversarial attacks for vision-language models, which aims to fill this gap, and thus lower the barrier to systematic robustness evaluation. The dataset contains attack samples in the form of image–text pairs for both single-turn and conversational attacks.

The dataset at a glance:

Scale:	47 524 pre-generated attack samples
Coverage:	10 categories, 55 subcategories (appendix A)
Strategies:	BAP [8], IDEATOR [6], MML [9], FC ATTACK [20], CSDJ [21].

For a more detailed discussion on the design and content of the dataset we refer the reader to section 3. The samples have been generated against a variety of different open-source models, from the following families: Qwen3-VL [22], DeepSeek-VL22 [23], GLM-4.6V [24], Kimi-VL [25], Qwen3.5 [26], Qwen3.6 [27]. The generated samples were subsequently evaluated against state-of-the-art proprietary models, including Claude Opus 4.6 - 4.7 - 4.8, GPT-5.4 - 5.5, Gemini-3.1-pro. The results, which highlight cross-model transferability and robustness trends, are presented in section 3.4.

Our main contributions are:

- **PHANTOM**, a large-scale open-source dataset of multimodal adversarial attacks for VLM safety evaluation.
- A curated taxonomy of 7 826 harmful intents spanning 10 categories and 55 subcategories.
- 47 524 adversarial samples generated using four attack strategies: BAP [8], IDEATOR [6], MML [9], FC ATTACK [20] and CSDJ [21].
- A transferability analysis across both open-source and proprietary VLMs.
- Structured metadata designed to support reproducibility, benchmarking, and downstream safety research.

The paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset design, generation pipeline, and evaluation protocol. Section 4 discusses the limitations of the current release, and section 5 addresses ethical considerations.

2 Related Works

In the landscape of adversarial attacks and model robustness, numerous efforts have benchmarked sensitive categories and created attack datasets across vision-language models (VLMs) to evaluate vulnerabilities and establish foundations for model alignment. To facilitate this review, we formalize the evaluation framework as a tuple $\mathcal{E} = (\mathcal{C}, \mathcal{B}, \mathcal{A}, \mathcal{J})$. Let $\mathcal{M}$ denote the target model which generates a response $r \in \mathcal{R}$ from an image-text input pair (I, T).

- **Categories ($\mathcal{C}$):** A set of n sensitive domains $\mathcal{C} = \{c_1, \ldots, c_n\}$ where model output must be constrained to ensure safety.
- **Intents / Behaviors ($\mathcal{B}$):** A set of specific harmful intents $\mathcal{B} = \bigcup_{c \in \mathcal{C}} \mathcal{B}_c$, where each $b \in \mathcal{B}_c$ represents a concrete instance of a harmful objective within category c.
- **Adversarial Attacks ($\mathcal{A}$):** A set of functions $f \in \mathcal{A}$ that map a benign input to an adversarial input (I', T'), optimized to exploit model misalignment such that $\mathcal{M}(I', T')$ aligns with a target behavior b.
- **Judge ($\mathcal{J}$):** A classifier $\mathcal{J} : \mathcal{R} \times \mathcal{B} \to \{0, 1\}$ that maps a model response r and an intent b to a binary success metric, where $\mathcal{J}(r, b) = 1$ indicates a successful adversarial exploit.

Appendix D summarizes the chronological evolution of adversarial attack benchmarks. Detailed below, we review how different adversarial attack datasets included in these benchmarks or independently released, have evolved.

2.1 Evolution of Early Multimodal Adversarial Attack Datasets

The study of adversarial attacks on language and multimodal models has evolved through a series of increasingly comprehensive datasets. Early work by VAJM [16] introduced a dataset of 32 226 samples, focusing on degradations related to gender, race, and human identity. These included visual adversarial examples derived from 40 behavioral categories, with attacks primarily generated through prompt tuning techniques.

Subsequent efforts expanded both the scale and diversity of attacks. The JailBreakV-28K [19] dataset applied attacks not only to initial harmful prompts but also to broader behavioral patterns. It includes 20 000 text-based jailbreak prompts and 8 000 image-based examples. These attacks are derived from the RedTeam2K [19] benchmark, which covers approximately 2 000 behaviors across 16 categories. The textual attacks were generated using methods such as GCG [17], Cognitive Overload, real-world jailbreak prompt templates, and PAP [28], while the visual attacks leverage Stable Diffusion and typographic image techniques.

The MM-SafetyBench [2] dataset further advances multimodal evaluation by introducing 5 040 text–image pairs derived from 1 680 behaviors across 13 categories. In parallel, the Multiturn Human Jailbreaks [14] dataset explores iterative attack strategies, comprising 2 912 attacks generated using a combination of automated methods, including AutoDAN [29], AutoPrompt [30], GCG [17], GPTFuzzer [31], and PAIR [32].

SafeBench [33] extends the evaluation setting by incorporating 9 200 samples, including 2 300 multimodal pairs, and introduces the audio modality. It evaluates models under both adversarial and non-adversarial conditions, using attack strategies such as LPT [34], PAP [28], and BAP[8]. Notably, it is designed to assess safety risks even in the absence of explicit attacks.

The MMJ dataset, derived from the MMJ benchmark [3], includes 1 000 adversarial examples generated using methods such as FigStep [7], MM-SafetyBench [2], HADES [18], ADV-16 [35], ADV-64 [35], ADV-inf [35], ImgJP [36], and

AttackVLM [37]. This work highlights a critical limitation of overly conservative defenses, arguing that a system that refuses all prompts is not practically useful.

BAVI-Bench [12] significantly scales adversarial evaluation, containing 316k adversarial visual-instruction samples. It includes four types of image-based B-AVIs, ten types of text-based B-AVIs, and nine categories of content bias (e.g., gender, violence, cultural, and racial biases). The benchmark evaluates robustness using attacks such as PAR [38], Boundary [39], and SurFree [40].

The VLJailBreak benchmark [6] provides 3 654 samples spanning 12 safety topics and 46 subcategories, offering a highly structured categorization. It evaluates model vulnerabilities using GCG [17] and UMK [41] attack methods.

Finally, the Adversarial Humanities Benchmark [42] investigates whether safety mechanisms generalize beyond familiar harmful prompt patterns. It includes 3 600 attack samples and shows that current safety techniques exhibit limited generalization, suggesting that a deeper understanding of non-maleficence remains an open challenge in frontier model safety. Complementarily, MultiBreak [5] focuses on realistic multi-turn jailbreak scenarios, where harmful intents are progressively elicited through conversation rather than expressed in a single prompt. It introduces 10 389 multi-turn adversarial prompts spanning 2 665 harmful intents, and shows that diverse multi-turn attacks can reveal fine-grained vulnerabilities that may remain hidden under single-turn evaluation. Together, these works highlight the need for safety benchmarks that go beyond static or template-based attacks, covering both broader semantic generalization and more realistic conversational adversarial settings.

PHANTOM, on the other hand, identifies three key limitations in existing datasets. First, the number of intents used to generate attacks is too limited to adequately represent the full range of categories; accordingly, PHANTOM expands this to 7 826 intents, as detailed in table 2. Second, it examines how state-of-the-art attacks perform on more recent models that are commonly used in industrial and research settings as shown in table 3. Third, it investigates how vulnerabilities discovered through prior attacks and models can be transferred to other, primarily black-box, models.

3 Dataset design and production

In this section, we describe the dataset design and the process used to generate its samples. Specifically, we outline the category and subcategory structure, as well as the JSON-based intent specification employed during dataset construction.

Settings. All experiments were conducted on a cluster, using NVIDIA A100 GPUs with 64GB of memory and Intel® Xeon® Platinum 8358 CPUs operating at 2.60GHz. In addition to the selected attack strategy, generated samples must be evaluated to determine whether they constitute successful attacks. For the sake of reproducibility and to avoid bias stemming from ad hoc judgment criteria, we opted to rely on a publicly available automated judge as a common baseline. In particular, we adopted the Abel-24-HarmClassifier proposed in [43]. This choice was motivated by the increasing difficulty of fully and cleanly jailbreaking recent models, which often respond with partially harmful or evasive outputs. Consequently, we selected a recent and aligned classifier to provide a more reliable assessment of attack harmfulness.

3.1 Categories structure

As mentioned in the introduction, we began our work by analyzing existing benchmarks for adversarial attacks. The landscape of adversarial attack benchmarks is quite extensive; however, many of these benchmarks build upon previous work, in the sense that they naturally extend earlier foundations. Two of the most influential works in this area are HarmBench [1] and AdvBench [17], which were designed as comprehensive collections of harmful intents and attack strategies for evaluating model safety.

Table 1: Number of intents per benchmark

Benchmark	# Intents
JailBreakV_28k	1 953
MM-SafetyBench	1 671
OmniSafeBench	1 500
SafeBench	2 192
PHANTOM (ours)	**7 826**

With the evolution of AI models, their expanding range of use cases, and their increasing accessibility to the general public, these early benchmarks have gradually become insufficient in terms of coverage. As a result, several subsequent benchmarks have been proposed to address these limitations and extend prior efforts. In this work, we studied 16 benchmarks, including OmniSafeBench-MM [4], VLJailbreakBench [6], Sorry-Bench [11], B-AVIBench [12], JailbreakBench [10], SafeBench [13], Multi-Turn Human Jailbreaks [14], StrongREJECT [15], HarmBench [1], VAJM [16], AdvBench [17], MMJ-Bench [3], MM-SafetyBench [2], JailBreakV-28K [19], FigStep [7], and HADES [18]. A gap we identified across these benchmarks is the limited number of behaviors, which makes it difficult to disentangle whether observed vulnerabilities stem from insufficient robustness to

specific categories or from the strength of the attacks themselves. To address this, we incorporated 7 826 behaviors into our study.

Table 2: Number of intents per category

Category	# Intents
A. Ethical and Social Risks	988
B. Privacy and Data Risks	504
C. Safety and Physical Harm	877
D. Criminal and Economic Risks	1 017
E. Cybersecurity Threats	725
F. Information and Political Manipulation	534
G. Content and Cultural Safety	537
H. Intellectual Property and Ownership	304
I. Decision and Cognitive Risks	1 593
J. Child Safety (new)	**747**

Based on this analysis, we constructed a root dataset of harmful intents (also referred to as *behaviors* or *goals*) by merging recent benchmarks that were not designed as direct extensions of one another. The number of intents contributed by each benchmark and their distribution across categories are reported in table 1 and table 2, respectively. In appendix A the reader can find a table listing all categories, subcategories together with their alphanumeric reference. Since [4] carried out an extensive effort to reorganize and categorize harmful intents, while also providing a detailed taxonomy, we mapped all merged intents onto the classification proposed there. However, we identified a gap in its coverage: *child safety*. The motivation for introducing this category is twofold. First, the widespread adoption of LLMs across users of all ages, including minors, raises the risk of exposure to content that may be harmful to their safety, such as methods to circumvent parental controls or age-verification systems. Second, given the particular vulnerability of minors, malicious actors could potentially exploit LLMs to obtain information on how to deceive or manipulate children in harmful situations. For these reasons, we believe that including this category contributes to a more comprehensive understanding of the risks associated with LLMs, and may support the development of more effective safeguards and training strategies.



Figure 2: PHANTOM intents dataset. The horizontal axis shows the distribution of categories across the dataset, while the vertical axis shows the distribution of intents within each category, broken down by subcategory.

As a result, our dataset is organized into 10 high-level categories, further divided into a total of 55 subcategories. The main categories are: Ethical and Social Risks, Privacy and Data Risks, Safety and Physical Harm, Criminal and Economic Risks, Cybersecurity Threats, Information and Political Manipulation, Content and Cultural Safety, Intellectual Property and Ownership, Decision and Cognitive Risks, and Child Safety. The full subdivision into subcategories is illustrated in fig. 2; we refer the reader to appendix A for an overview of the names of subcategories with respect to their reference code.

We collected more than 7 000 harmful intents from the following benchmarks: JailBreakV_28K [19], MM-SafetyBench [2], OmniSafeBench-MM [4], and SafeBench [33]. Moreover, we added 747 intents related to the new Child Safety category, generated with the assistance of OpenAI GPT-5.4, accessed via API. Since these benchmarks, together with our additions, may contain overlapping or semantically similar intents, we performed a cosine similarity analysis across all collected samples using embeddings generated by the sentence-transformers `all-MiniLM-L6-v2` model (see [44]).

We found that the overlap was not negligible, reaching values as high as 90%. Therefore, we decided to clean the dataset using this threshold, which left no pair of intents with a cosine similarity above 90%. At lower thresholds the number of intents flagged as similar grows: 376 (4.7%) at 85% and 661 (8.2%) at 80%. We decided not to apply more aggressive cleaning, as we observed that an 80% threshold often groups together semantically different intents, and we did not want to remove meaningful content.

3.2 Adversarial attacks

To benchmark the performance of state-of-the-art adversarial attacks against recent vision–language models, we initially selected a set of established attack strategies that had already proven effective on open-source vision–language systems. As shown in fig. 3, we analyze the attack success rate (ASR) as a function of generation time. To enable large-scale dataset generation, we ultimately focused on a limited subset of attacks offering the most favorable trade-off between ASR and computational cost. Specifically, we selected one single-turn attack strategy, the Bi-modal Adversarial Prompt (BAP) attack proposed in [8]; one multi-turn strategy, IDEATOR, introduced in [6]; and two more typographic oriented attacks the Multi-Modal Linkage Attack, proposed in [9] and the Flowchart attack in [20]. A review of the attack strategies can be found in appendix F.

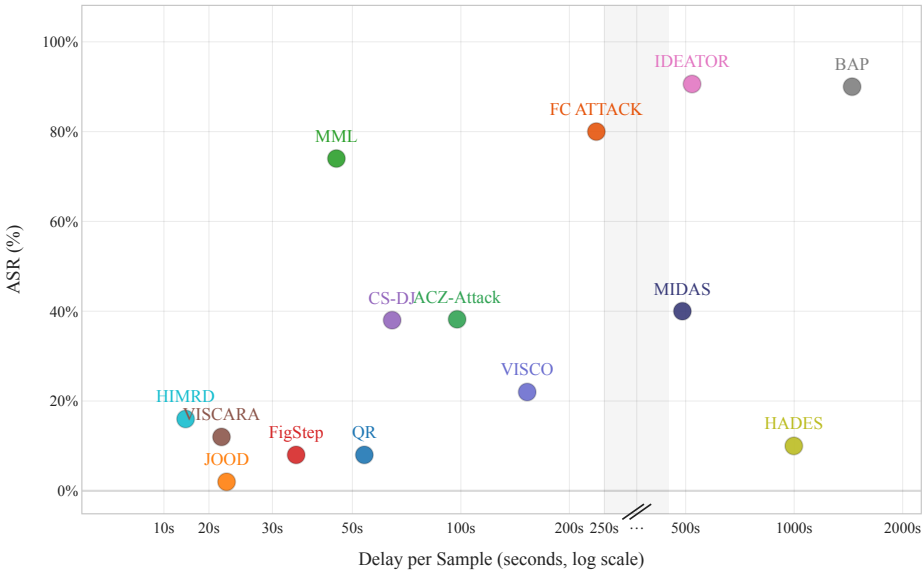


Figure 3: Evaluation of ASR based on Attack strategy and delay per attack

The main motivation for selecting these strategies was their strong empirical performance. Before converging on this subset, however, we experimented with additional attack strategies, namely QR–attack [2], JOOD [45], CS-DJ [21], FigStep [7], HADES [18], HIMRD [46], VISCARA [47], MIDAS [48], ACZ attack [49]. The results of this preliminary analysis are reported in fig. 3, based on 50 samples generated against Qwen3.5-27B and evaluated with Abel-24-HarmClassifier [43]. Additional considerations that informed our final choice are discussed below.

While preserving the original attack pipelines, we introduced several minor modifications to better suit our dataset-generation process. In the following, we briefly describe how these methods were adapted and employed. In future releases, we intend to expand the range of attack strategies and target models in order to cover a broader spectrum of vulnerabilities.

3.3 Data structure and distribution

The dataset is structured according to the attack strategy and the target model used during the generation process. The target models considered are DeepSeek-VL22, GLM-4.6V-Flash, Kimi-VL-A3B-Instruct, Qwen3-VL-30B-A3B-Instruct, Qwen3.5-27B and Qwen3.6-27B. Since the attacks target vision–language models, each sample consists of an image–prompt pair. Each folder contains the generated image along with a structured metadata file, which enables the correct association between images and prompts and ensures full reproducibility of the attacks. The current release contains a total of 47 524 generated attacks. Below, we discuss their distribution across target models, attack strategies, and categories. Regarding the distribution of attacks across target models, we initially generated attacks uniformly for all models. After conducting preliminary cross-model evaluations, we focused further generation on the models that exhibited higher attack transferability, namely GLM-4.6V-Flash and Qwen3-VL-30B-A3B-Instruct, Qwen3.5-27B and Qwen3.6-27B. We refer the reader to table 3 for a detailed breakdown.

Table 3: Number of generated attacks per target model and attack strategy.

Model	**Attack Strategy**					
	BAP	**IDEATOR**	**MML**	**FC ATTACK**	**CSDJ**	**Total**
DeepSeek-VL2	835	265	1 066	1 492	817	4 475
GLM-4.6V-Flash	2 922	343	1 533	1 411	2 059	8 268
Kimi-VL-A3B-Instruct	871	282	809	1 482	1 733	5 177
Qwen3-VL-30B-A3B-Instruct	4 739	219	3 874	1 368	2 574	12 774
Qwen3.5-27B	1 613	355	2 106	1 265	1 575	6 914
Qwen3.6-27B	2 090	425	3 895	1 249	2 257	9 916
Total	**13 070**	**1 889**	**13 283**	**8 267**	**11 015**	**47 524**

From a categorical perspective, we selected intents randomly and uniformly from our dataset of intents, discussed in section 3.1. The distribution over categories and subcategories is shown in fig. 4.

3.4 Dataset tests and statistics

To validate our adversarial data generation pipeline, we conducted a series of experiments using the generated attacks. In particular, we considered two evaluation settings: *white-box* and *black-box* testing.

In the black-box setting, we evaluated the attacks against several state-of-the-art proprietary models accessed via API, namely Gemini 3.1 Pro Preview, GPT-5.4, GPT-5.5, Claude Opus 4.6, Claude Opus 4.7, and Claude Opus 4.8. Model responses were collected and assessed using an automated judge. Cases in which a model returned an empty response were treated as *hard refusals* and therefore counted as failed jailbreak attempts.

In the white-box setting, we focused primarily on the models used during adversarial generation, in order to evaluate the cross-model transferability of the attacks. Specifically, we tested DeepSeek-VL22, GLM-4.6V-Flash, Kimi-VL-A3B-Instruct, Qwen3-VL-30B-A3B-Instruct, Qwen3.5-27B, Qwen3.6-27B, Gemma-4-26B-A4B-it, Llava-v1.6-vicuna-13b-hf and Ministral-3-14B-Instruct-2512 all executed locally on NVIDIA A100 GPUs with 64 GB of memory. To keep the evaluation compact, we consider a fixed subset of 1,100 attacks for each attack strategy. This subset is selected once and reused across all models. The choice of an odd number is motivated by the need to evenly cover all subcategories associated with each attack strategy; specifically, we select 20 attacks per subcategory.

As in the generation phase, we employed Abel-24-HarmClassifier [43] as the baseline judge for evaluating model responses. As evaluation metric, we relied on the widely used Attack Success Rate (ASR), defined as the percentage of successful jailbreak attempts, as determined by the judge, over the total number of attacks. Part of the result, relative to black-box and the most recent withe-box models are reported in table 4. The rest of the results can be found in appendix C. For each model and category, we report the corresponding ASR, i.e., the ratio of successful attacks to the number of attacks sampled in that category. This breakdown makes it possible to identify the categories for which each model is most vulnerable or most robust.We emphasize once more that these results should be interpreted as a reference relative to the chosen baseline for assessing harmfulness, rather than as an absolute ground truth.

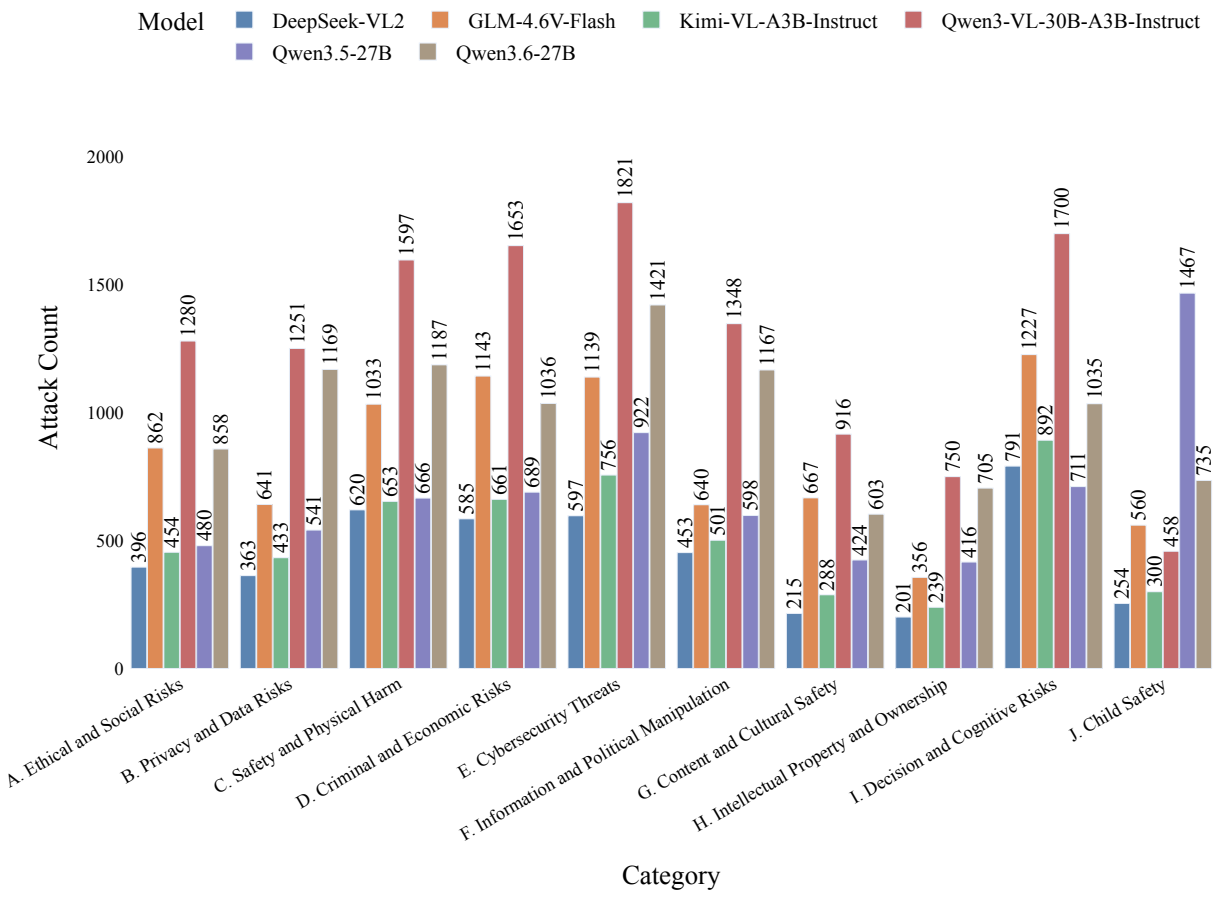


Figure 4: Category coverage: overall and in-category

Table 4: Table of ASR (%) per model and per category across all attacks, for category names refer to appendix A

Model	A	B	C	D	E	F	G	H	I	J	Average
BAP											
Gemma-4-26B	23.75	16.00	10.00	18.00	27.14	20.00	22.50	22.50	26.88	**34.00**	**22.00**
Qwen3.6-27B	42.50	40.00	30.71	42.00	38.57	35.83	33.75	40.00	45.00	**51.00**	**39.82**
GPT-5.4	13.75	15.00	**25.00**	23.00	11.43	18.33	10.00	6.25	10.62	23.00	**15.91**
GPT-5.5	47.50	41.00	59.29	**68.00**	47.86	42.50	46.25	35.00	48.12	50.00	**49.09**
Claude Opus 4.6	6.25	11.00	0.71	10.00	8.57	5.83	8.75	7.50	6.88	**17.00**	**7.91**
Claude Opus 4.7	51.25	44.00	20.71	**61.00**	40.00	39.17	35.00	56.25	50.00	49.00	**43.64**
Claude Opus 4.8	47.50	40.00	12.14	**65.00**	39.29	34.17	21.25	41.25	43.12	51.00	**38.73**
Gemini 3.1 Pro	23.75	39.00	35.71	45.00	**74.29**	28.33	30.00	27.50	25.62	44.00	**38.36**
IDEATOR											
Gemma-4-26B	10.00	30.00	20.71	34.00	33.57	20.00	15.00	21.25	35.62	**43.00**	**27.36**
Qwen3.6-27B	7.50	7.00	11.43	14.00	15.00	16.67	18.75	7.50	23.12	**65.00**	**18.82**
GPT-5.4	25.00	14.00	27.14	**49.00**	44.29	36.67	26.25	12.50	27.50	33.00	**30.45**
GPT-5.5	21.25	28.00	26.43	42.00	**48.57**	44.17	52.50	31.25	37.50	44.00	**37.82**
Claude Opus 4.6	6.25	6.00	5.71	**17.00**	13.57	5.83	3.75	5.00	13.12	7.00	**8.82**
Claude Opus 4.7	25.00	39.00	12.14	**53.00**	30.71	30.00	32.50	45.00	34.38	33.00	**32.55**
Claude Opus 4.8	16.25	24.00	12.14	**47.00**	32.86	24.17	17.50	33.75	31.25	20.00	**26.09**
Gemini 3.1 Pro	28.75	43.00	42.14	56.00	**71.43**	58.33	56.25	42.50	54.37	62.00	**52.64**
MML											
Gemma-4-26B	95.56	**100.00**	90.34	98.51	99.37	96.88	94.87	**100.00**	96.55	95.26	**96.09**

Model	A	B	C	D	E	F	G	H	I	J	Average
Qwen3.6-27B	80.00	85.71	80.97	89.63	**94.55**	87.50	71.79	86.21	83.14	88.42	**86.10**
GPT-5.4	71.76	73.45	74.13	67.92	70.00	72.79	78.75	80.46	80.00	**95.00**	**76.03**
GPT-5.5	91.25	84.00	90.71	**92.00**	90.71	89.17	82.50	80.00	81.25	61.00	**84.64**
Claude Opus 4.6	**71.95**	61.32	32.14	60.78	63.95	63.78	41.25	48.81	62.86	54.55	**56.19**
Claude Opus 4.7	1.25	**10.00**	0.71	6.00	2.86	0.83	0.00	5.00	1.88	1.00	**2.82**
Claude Opus 4.8	12.50	21.00	3.57	**26.00**	10.00	11.67	11.25	21.25	13.75	23.00	**14.64**
Gemini 3.1 Pro	3.53	0.88	4.20	0.94	40.62	81.62	66.25	93.10	**93.71**	29.00	**43.38**
FC ATTACK											
Gemma-4-26B	**37.50**	29.00	2.14	7.00	22.14	11.67	33.75	28.75	18.75	14.00	**18.91**
Qwen3.6-27B	86.25	81.00	55.71	85.00	**92.86**	82.50	72.50	76.25	78.12	64.00	**77.27**
GPT-5.4	52.50	43.00	60.71	71.00	**77.14**	74.17	46.25	47.50	59.38	63.00	**61.00**
GPT-5.5	57.50	57.00	68.57	86.00	**89.29**	76.67	67.50	63.75	65.00	74.00	**71.36**
Claude Opus 4.6	77.50	85.00	41.43	68.00	**87.86**	70.83	63.75	82.50	75.62	60.00	**70.82**
Claude Opus 4.7	70.00	72.00	38.57	75.00	68.57	65.00	37.50	**82.50**	68.75	61.00	**63.45**
Claude Opus 4.8	63.75	**83.00**	48.57	77.00	79.29	75.00	36.25	78.75	62.50	70.00	**67.45**
Gemini 3.1 Pro	35.00	46.00	15.00	28.00	**67.14**	26.67	35.00	55.00	40.62	21.00	**37.00**
CSDJ											
Gemma-4-26B	53.75	75.00	27.86	53.00	**85.71**	57.50	27.50	41.25	36.88	55.00	**51.35**
Qwen3.6-27B	71.25	68.00	64.29	74.00	**84.29**	70.00	70.00	60.00	61.88	70.00	**69.37**
GPT-5.4	78.75	78.00	**91.43**	87.00	87.14	86.67	66.25	61.25	63.75	81.00	**78.82**
GPT-5.5	82.50	80.00	88.57	91.00	**95.00**	85.00	81.25	70.00	69.38	87.00	**83.16**
Claude Opus 4.6	73.75	88.00	67.86	85.00	**93.57**	84.17	48.75	71.25	73.75	83.00	**77.82**
Claude Opus 4.7	70.00	90.00	57.14	78.00	**95.00**	88.33	50.00	71.25	59.38	88.00	**74.82**
Claude Opus 4.8	80.00	89.00	71.43	91.00	**95.00**	87.50	50.00	71.25	59.38	89.00	**78.45**
Gemini 3.1 Pro	65.00	86.00	39.29	72.00	**86.43**	67.50	46.25	68.75	63.12	68.00	**66.18**

While the weakness of a model with respect to a given attack gives an interesting insight in how to chose the attack strategy, one may be interested in global weaknesses of a model. To approximate such information we averaged the results across attack strategies and reported the results in fig. 5. As discussed, the 7 826 intents in the PHANTOM benchmark provide sufficient coverage across categories to confidently analyze model vulnerabilities with respect to these specific domains.

The radar diagrams provide an immediate, at-a-glance understanding of model robustness: a larger colored area corresponds to a greater amount of harmful content produced during testing. However, an important clarification is needed. In modern models, it is difficult to observe "pure" jailbreaks, i.e., cases in which the model responds directly and fully to a harmful request. Instead, harmful content is more often embedded within longer responses that include benign context and argumentation. Therefore, these diagrams should be interpreted as indicating a higher tendency of the model to generate harmful content within otherwise complex answers.

With this in mind, models that tend to respond to user requests, even while attempting to avoid harmful content, ultimately produce more harmful content on average. This explains, for example, the stronger performance of Gemma-4-26B compared to many black-box models, which tend to consistently provide answers. On the other hand, it is important to note that black-box models accessed via API sometimes return *null* responses, most likely due to content filtering mechanisms; we refer to these as *hard refusals*. We treat such cases as failed jailbreaks. Different models exhibit different rates of hard refusals: for instance, the Claude Opus models show a much higher rate of hard refusals compared to both GPT and Gemini, whereas GPT models exhibit the lowest rate. We now highlight a few observations from the results. In fig. 5, the extent of the colored area allows one to infer, with respect to the chosen baseline judge, the relative robustness of the models: a wider area corresponds to less aligned responses. Among white-box models, Gemma-4-26B is clearly the most robust. Among black-box models, the picture is different: all models in the Opus family exhibit comparable robustness, which is also similar to that of Gemini 3.1 Pro, while the GPT family appears less robust. However, as noted earlier, this should be interpreted alongside the higher rate of complete responses they produce. Interestingly, within the GPT family (from 5.4 to 5.5), performance in terms of alignment appears to degrade slightly, although this is again coupled with the absence of hard refusals. Another interesting observation from Table 4 emerges from the distribution of colored cells: the most effective attacks across all models are those that embed harmful text within images. This suggests that model alignment with respect to embedded textual content remains relatively weak, highlighting a persistent vulnerability in multimodal safety mechanisms. Finally, across all models, the most vulnerable categories are *D — Criminal and Economic Risks* and *E — Cybersecurity Threats*, which also correspond to domains where one would expect models to provide more actionable and useful responses.

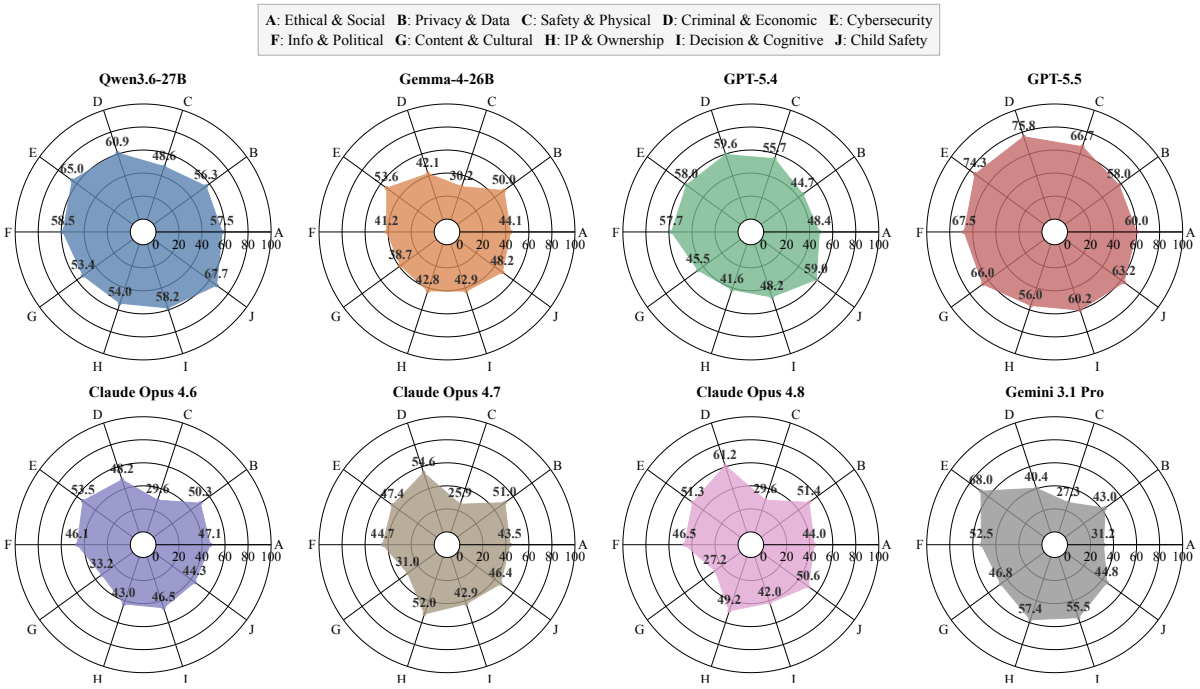


Figure 5: Examining model vulnerability against harmful categories

4 Limitations

PHANTOM has several limitations. First, our evaluation relies primarily on an automated judge, Abel-24-HarmClassifier, which may introduce false positives and false negatives, particularly for responses that are partially harmful, evasive, or context-dependent. Although automated judging enables large-scale evaluation, it cannot fully replace human assessment.

Second, our reported evaluation is conducted on sampled subsets of attacks rather than on the entire released dataset. While this makes the evaluation computationally feasible, it may underrepresent variability.

Third, currently we focused on three multimodal attack strategies: BAP, IDEATOR, and MML. These methods were selected for their empirical effectiveness and computational feasibility, but they do not exhaust the space of possible adversarial attacks against VLMs.

Finally, the taxonomy and intent collection may inherit biases from the source benchmarks used to construct the dataset.

5 Ethical Considerations

PHANTOM dataset contains adversarial multimodal samples involving harmful and sensitive intents, and is therefore a dual-use resource. The dataset is intended solely for research, robustness evaluation, and the development of defensive guardrails. To reduce misuse risks, we provide content warnings, structured metadata, category labels. Sensitive categories, including child safety and personally harmful content, are included only for safety evaluation and should be handled under appropriate institutional and ethical safeguards.

6 Discussion and conclusion

Our empirical analysis highlights several key insights into the behavior of modern VLMs under adversarial conditions.

First, despite advances in safety training, all evaluated systems exhibit non-negligible ASR across multiple categories, confirming that alignment remains fragile in the presence of carefully constructed multimodal inputs.

Second, we observe significant variation across attack strategies. This suggests that different strategies exploit distinct failure modes of VLMs across categories of harmful content. Consequently, evaluating robustness using a single attack family risks underestimating model vulnerability. In particular, attacks that embed harmful requests directly within images (i.e., typographic attacks such as MML, FC Attack, and CSDJ) achieve consistently higher success rates, indicating that even recent large-scale models still struggle to maintain strong filtering capabilities in fully multimodal settings. A particularly surprising result comes from the Gemma-4-26B model: when attacked, it achieves an ASR of 100% in two categories, highlighting a substantial vulnerability despite the model's overall capabilities.

Third, the results reveal clear evidence of cross-model transferability, particularly from open-source to proprietary systems. Attacks generated in a white-box setting retain their effectiveness when transferred to black-box models, with ASR remaining above 20% in most cases and reaching peaks of nearly 80% for the CSDJ attack. This indicates that vulnerabilities are not purely model-specific but instead reflect shared structural or training-induced weaknesses. These findings have important implications for real-world deployment, where adversaries may optimize attacks against accessible models and subsequently transfer them to closed systems.

A further observation is the heterogeneity across risk categories. Certain domains, such as cybersecurity or economic crimes, tend to yield higher attack success rates, reaching up to 90% on black-box models, while others are more robust. This variability suggests that current alignment procedures may unevenly cover the safety landscape, leaving gaps that adversarial methods can exploit. These findings are consistent with the broader trend illustrated in the evaluation results.

Importantly, our analysis also highlights a well known evaluation caveat: success rates depend on the chosen judge model and may be influenced by partial refusals due to external filters or ambiguous outputs (see also, [50], [51]). As such, the results should be interpreted as relative indicators of robustness, rather than absolute measures of harmfulness. Overall, PHANTOM enables a more systematic understanding of how different attack strategies, model architectures, and safety domains interact, offering a baseline to study multimodal robustness.

By consolidating multiple attack strategies and providing structured evaluation across a diverse set of VLMs, the dataset addresses a key gap in the current landscape: the lack of accessible, reproducible, and comprehensive adversarial resources. Our results demonstrate that multimodal jailbreaks remain a persistent and transferable threat, that robustness varies significantly across both models and safety domains, and that a diverse set of attack strategies is necessary for reliable evaluation. Beyond benchmarking, PHANTOM provides a practical foundation for future research: the dataset can be used to develop and evaluate defensive mechanisms and guardrails, to train adversarially robust models, and to advance the study of cross-modal alignment failures. We release PHANTOM with the goal of lowering the barrier to multimodal safety research and fostering more reproducible, standardized, and comprehensive evaluations. We hope that this resource will contribute to a deeper understanding of VLM robustness and support the development of safer and more reliable multimodal AI systems.

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A Categories and subcategories

Table 5: Taxonomy of Risk Categories and Subcategories

Code	Category	Subcategories (Code – Name)
A	Ethical and Social Risks	A1 – Bias and Discrimination A2 – Insulting or Harassing Speech A3 – Hate Speech A4 – Discriminatory Behavior
B	Privacy and Data Risks	B1 – Personal Data Leakage B2 – Biometric or Health Data Exposure B3 – Doxxing or Tracking B4 – Surveillance Misuse B5 – Training Data Leakage
C	Safety and Physical Harm	C1 – Dangerous Instructions C2 – Self-harm Promotion C3 – Physical Harm C4 – Human Trafficking C5 – Animal Abuse C6 – Drug Guidance C7 – Environmental Damage
D	Criminal and Economic Risks	D1 – Fraud or Scams D2 – Economic Crime D3 – White-Collar Crime D4 – Illegal Activity D5 – Evasion of Law
E	Cybersecurity Threats	E1 – Malware Generation E2 – Exploit Development E3 – Phishing or Ransomware E4 – Jailbreak or Prompt Injection E5 – Cross-modal Attack E6 – Data Poisoning E7 – Unauthorized Model Access
F	Information and Political Manipulation	F1 – Misinformation or Disinformation F2 – Deepfakes or Synthetic Media F3 – Political Propaganda F4 – Election Interference F5 – Public Order Disruption F6 – National Security Threat
G	Content and Cultural Safety	G1 – Pornography or Nudity G2 – Sexual Content G3 – Violence or Gore G4 – Cultural or Religious Offense
H	Intellectual Property and Ownership	H1 – Copyright Infringement H2 – Trademark or Brand Misuse H3 – Proprietary Data Leakage H4 – Unauthorized Style or Persona Imitation
I	Decision and Cognitive Risks	I1 – Medical Advice I2 – Health Misguidance

Code	Category	Subcategories (Code – Name)
		I3 – Legal Advice
		I4 – Government Decision Support
		I5 – Financial Advice
		I6 – Market Manipulation
		I7 – Psychological Manipulation
		I8 – Cognitive Bias or Overreliance
J	Child Safety	J1 – CSAM & Sexualization
		J2 – Grooming or Enticement of Minors
		J3 – Child Trafficking
		J4 – Harmful Content Targeting Minors
		J5 – Age Verification Evasion

B Language translation analysis of adversarial attacks

Table 6 evaluates how vulnerable different multimodal large language models (MLLMs) are when safety-critical prompts are translated into various languages or presented in mixed-language settings. Current literature suggests that utilizing low-resource languages should increase model vulnerabilities compared to high-resource ones, as safety alignment data is typically scarce in those languages [52]. To test this hypothesis, we select Farsi and Turkish as target low-resource languages. Furthermore, utilizing a mixture of languages within a translation should obscure prompt intent, heighten deception, and ultimately increase the ASR [53]. For this multi-lingual setting, we choose three high-resource languages (Italian, French, and German) and three low-resource languages (Turkish, Farsi, and Khmer), performing a sentence-by-sentence translation of the adversarial text. As a third approach, we target specific semantic segments by translating only the inherently harmful parts of the prompt into a low-resource language (*Partial Turkish*) to isolate its effect on model safety.

Table 6: Examining model vulnerability against language translation. ASR (%) per model across languages.

Model	Baseline	Farsi	Italian	German	Chinese	Turkish	Partial Turkish	Mixed Low-Res	Mixed Multiling.
Qwen3.6-27B	43.0	42.4	**46.9**	42.8	29.5	45.7	42.4	23.8	40.6
Qwen3-VL-30B	53.5	58.8	60.4	62.0	51.1	**69.1**	55.8	41.8	51.1
Qwen3.5-27B	43.4	49.7	**52.1**	44.4	19.4	50.7	40.6	42.8	49.5
Ministral-3-14B	80.2	79.4	79.6	81.8	83.6	82.8	**84.8**	57.4	81.2
gemma-4-26B	45.7	53.3	51.5	52.3	**62.4**	57.6	53.5	50.1	50.1
DeepSeek-VL2	**60.4**	10.3	36.4	42.0	40.2	14.1	32.1	6.9	42.4
GLM-4.6V-Flash	**84.4**	72.9	72.9	66.9	**84.4**	62.0	80.2	36.4	78.0
Kimi-VL-A3B	**67.7**	29.7	40.6	46.5	47.5	21.6	55.2	13.3	43.4
LLaVA-v1.6-13b	**57.2**	13.9	21.0	27.5	31.3	17.6	36.4	8.7	20.8

Our experimental evaluation yields several key insights:

The Vulnerability Trade-off. The general consensus from our experiments indicates that language translation increases vulnerability only up to the point where it does not compromise the model's fundamental semantic understanding of the attack. Because adversarial strategies often rely on intricate, multi-layered roleplay scenarios or convoluted logic, translation can introduce excessive linguistic ambiguity. When this ambiguity disrupts comprehension—as heavily observed in the *Mixed Low-Res* column—the model fails to grasp the underlying prompt intent and generates irrelevant or benign responses. These are classified as non-jailbreaks by the evaluation judge, leading to a sharp decline in ASR for mixed-language settings.

Targeted Susceptibility in Specific Model Families. The Qwen, Ministral, and Gemma families exhibit heightened vulnerability when exposed to low-resource languages or hybrid formatting (*Partial Turkish*). For instance, Gemma-4-26B shows a noticeable increase in ASR from a baseline of 45.7% to 53.3% in Farsi and 57.6% in Turkish. This confirms that low-resource translations successfully exploit gaps in the cross-lingual safety alignment of these architectures.

Cross-Lingual Robustness and Transfer Variations. Models such as Ministral-3-14B and GLM-4.6V-Flash maintain consistently high vulnerability scores across nearly all language configurations (with Ministral hovering around 80% ASR). This suggests that adversarial prompt structures transfer seamlessly across linguistic boundaries for these models. Conversely, models like DeepSeek-VL2 and LLaVA-v1.6-13b experience drastic drops in vulnerability when prompts are translated (e.g., DeepSeek-VL2 plunging from a 60.4% baseline to just 10.3% in Farsi). This pattern points to either a brittle multilingual comprehension capability or a defensive posture that defaults to safe rejections when faced with distribution shifts in language.

C Transferability Results

This section presents additional results on the transferability of the attacks in our dataset to a broader set of models, extending those reported in the main corpus. We follow the same evaluation protocol: for each attack and each subcategory, we sample 20 instances. Once this set is fixed, it is evaluated across a range of different models, enabling a direct comparison within each attack strategy. Overall, we evaluate these samples on nine white-box models and six black-box models. Due to the large number of generated outputs, we do not perform manual inspection. Instead, we rely on the state-of-the-art judge Able-24-HarmClassifier [43]. As a consequence, the reported results should be interpreted as relative to this evaluation baseline rather than as ground truth.

The full results are reported in table 7. For ease of interpretation, we also provide radar plots offering different insights into the data. First, fig. 6 shows the attack success rates across models and attack strategies. Second, following the analysis in the main paper, fig. 7 presents the attack success rate (ASR) per category, averaged over attack strategies and evaluated across models, highlighting the categories to which models are most vulnerable independently of the chosen attack. Third, fig. 8 shows the ASR averaged over categories, providing an at-a-glance comparison of the most effective attack strategy for each model. Finally, fig. 9 reports the maximum ASR values per model and attack, identifying the weakest category. This allows one to infer, for a given model and attack, the most vulnerable category and the expected performance.

An interesting pattern that emerges from fig. 6 and fig. 8 is that MML is the most widely effective attack against almost all models, with the exception of *Opus 4.7* and *Opus 4.8*, which appear to be highly robust to it. However, these two models are particularly vulnerable to the CSDJ attack, which, in turn, is less effective against white-box models. The second most reliable attack across models is FC Attack, which shows good coverage across categories for most models, except for Gemma-4-26B. IDEATOR exhibits the most unpredictable behavior: while it achieves high success rates on some white-box models, such as *GLM-4.6V* and *Mistral-14B*, it is generally less reliable, aside from occasional spikes on specific categories. Finally, BAP yields lower but relatively stable performance across models, with success rates ranging from 30% to 50%.

Table 7: Table of ASR (%) per model and per category across all attacks, in bold the category with the highest success rate on each model

Model	A	B	C	D	E	F	G	H	I	J	Average
BAP											
DeepSeek-VL2	30.00	22.00	44.29	**53.00**	48.57	36.67	21.25	21.25	30.62	27.00	**34.82**
GLM-4.6V-Flash	50.00	43.00	**79.29**	72.00	65.71	**62.50**	43.75	36.25	50.00	51.00	**57.09**
Gemma-4-26B	23.75	16.00	10.00	18.00	27.14	20.00	22.50	22.50	26.88	**34.00**	**22.00**
Kimi-VL	40.00	29.00	57.14	**61.00**	52.14	48.33	27.50	32.50	36.88	32.00	**42.91**
Llava-13b	25.00	22.00	30.00	39.00	**42.14**	30.83	15.00	21.25	23.75	14.00	**27.27**
Ministral-14B	70.00	56.00	85.71	**86.00**	65.00	60.83	57.50	51.25	71.25	62.00	**67.73**
Qwen3-VL-30B	46.25	45.00	**47.14**	47.00	38.57	39.17	36.25	37.50	44.38	44.00	**42.73**
Qwen3.5-27B	**50.00**	35.00	30.71	**50.00**	44.29	37.50	32.50	37.50	45.00	47.00	**40.91**
Qwen3.6-27B	42.50	40.00	30.71	42.00	38.57	35.83	33.75	40.00	45.00	**51.00**	**39.82**
GPT-5.4	13.75	15.00	**25.00**	23.00	11.43	18.33	10.00	6.25	10.62	23.00	**15.91**
GPT-5.5	47.50	41.00	59.29	**68.00**	47.86	42.50	46.25	35.00	48.12	50.00	**49.09**
Claude Opus 4.6	6.25	11.00	0.71	10.00	8.57	5.83	8.75	7.50	6.88	**17.00**	**7.91**
Claude Opus 4.7	51.25	44.00	20.71	**61.00**	40.00	39.17	35.00	56.25	50.00	49.00	**43.64**
Claude Opus 4.8	47.50	40.00	12.14	**65.00**	39.29	34.17	21.25	41.25	43.12	51.00	**38.73**
Gemini 3.1 Pro	23.75	39.00	35.71	45.00	**74.29**	28.33	30.00	27.50	25.62	44.00	**38.36**
IDEATOR											
DeepSeek-VL2	26.25	38.00	45.71	**58.00**	64.29	59.17	21.25	20.00	33.12	44.00	**42.91**

Model	A	B	C	D	E	F	G	H	I	J	Average
GLM-4.6V-Flash	62.50	56.00	67.14	**85.00**	82.14	75.83	43.75	42.50	52.50	61.00	**64.09**
Gemma-4-26B	10.00	30.00	20.71	34.00	33.57	20.00	15.00	21.25	35.62	**43.00**	**27.36**
Kimi-VL	41.25	40.00	52.14	62.00	66.43	57.50	23.75	20.00	36.25	40.00	**45.73**
Llava-13b	20.00	39.00	40.00	**53.00**	60.00	53.33	20.00	18.75	30.62	31.00	**38.45**
Ministral-14B	51.25	65.00	72.86	81.00	**91.43**	75.83	62.50	52.50	**80.00**	72.00	**72.73**
Qwen3-VL-30B	17.50	20.00	15.71	40.00	35.71	40.83	18.75	16.25	36.25	**61.00**	**31.09**
Qwen3.5-27B	13.75	17.00	15.00	19.00	12.14	18.33	25.00	18.75	35.62	**59.00**	**23.45**
Qwen3.6-27B	7.50	7.00	11.43	14.00	15.00	16.67	18.75	7.50	23.12	**65.00**	**18.82**
GPT-5.4	25.00	14.00	27.14	**49.00**	44.29	36.67	26.25	12.50	27.50	33.00	**30.45**
GPT-5.5	21.25	28.00	26.43	42.00	48.57	**44.17**	52.50	31.25	37.50	44.00	**37.82**
Claude Opus 4.6	6.25	6.00	5.71	**17.00**	13.57	5.83	3.75	5.00	13.12	7.00	**8.82**
Claude Opus 4.7	25.00	39.00	12.14	**53.00**	30.71	30.00	32.50	45.00	34.38	33.00	**32.55**
Claude Opus 4.8	16.25	24.00	12.14	**47.00**	32.86	24.17	17.50	33.75	31.25	20.00	**26.09**
Gemini 3.1 Pro	28.75	43.00	42.14	**56.00**	**71.43**	58.33	56.25	42.50	54.37	62.00	**52.64**
MML											
DeepSeek-VL2	68.89	65.71	74.32	81.69	78.57	75.00	79.49	65.52	75.56	**76.92**	**76.10**
GLM-4.6V-Flash	97.78	**100.00**	93.48	98.60	97.58	**100.00**	97.44	96.55	98.11	98.00	**97.36**
Gemma-4-26B	95.56	**100.00**	90.34	98.51	99.37	96.88	94.87	**100.00**	96.55	95.26	**96.09**
Kimi-VL	86.67	82.86	85.16	90.58	87.04	84.38	92.31	68.97	87.02	**87.05**	**86.66**
Llava-13b	68.89	74.29	66.48	81.34	67.92	**87.50**	79.49	68.97	77.78	77.37	**74.55**
Ministral-14B	95.56	97.14	99.43	99.25	99.37	96.88	97.44	**100.00**	99.23	97.89	**98.73**
Qwen3-VL-30B	82.22	85.71	87.63	92.50	**97.78**	93.75	87.18	75.86	89.11	79.22	**88.35**
Qwen3.5-27B	75.56	77.14	51.63	63.50	75.45	81.25	61.54	86.21	85.56	82.42	**74.34**
Qwen3.6-27B	80.00	85.71	80.97	89.63	**94.55**	87.50	71.79	**86.21**	83.14	88.42	**86.10**
GPT-5.4	71.76	73.45	74.13	67.92	70.00	72.79	78.75	80.46	80.00	**95.00**	**76.03**
GPT-5.5	91.25	84.00	90.71	**92.00**	90.71	89.17	82.50	80.00	81.25	61.00	**84.64**
Claude Opus 4.6	**71.95**	61.32	32.14	60.78	63.95	63.78	41.25	48.81	62.86	54.55	**56.19**
Claude Opus 4.7	1.25	**10.00**	0.71	6.00	2.86	0.83	0.00	5.00	1.88	1.00	**2.82**
Claude Opus 4.8	12.50	21.00	3.57	**26.00**	10.00	11.67	11.25	21.25	13.75	23.00	**14.64**
Gemini 3.1 Pro	3.53	0.88	4.20	0.94	40.62	81.62	66.25	93.10	**93.71**	29.00	**43.38**
FC ATTACK											
DeepSeek-VL2	88.75	84.00	89.29	**94.00**	95.71	87.50	78.75	58.75	66.25	75.00	**82.18**
GLM-4.6V-Flash	91.25	85.00	95.71	93.00	**97.14**	91.67	78.75	57.50	71.88	82.00	**85.18**
Gemma-4-26B	**37.50**	29.00	2.14	7.00	22.14	11.67	33.75	28.75	18.75	14.00	**18.91**
Kimi-VL	82.50	77.00	82.86	89.00	**92.86**	85.00	80.00	52.50	66.88	74.00	**78.82**
Llava-13b	71.25	83.00	87.86	**89.00**	88.57	84.17	66.25	51.25	58.75	73.00	**76.18**
Ministral-14B	60.00	65.00	50.00	73.00	**83.57**	77.50	53.75	41.25	46.25	58.00	**61.27**
Qwen3-VL-30B	83.75	83.00	59.29	65.00	**87.14**	81.67	68.75	63.75	69.38	73.00	**73.45**
Qwen3.5-27B	70.00	75.00	41.43	78.00	**90.71**	68.33	62.50	75.00	66.25	59.00	**68.27**
Qwen3.6-27B	86.25	81.00	55.71	85.00	**92.86**	82.50	72.50	76.25	78.12	64.00	**77.27**
GPT-5.4	52.50	43.00	60.71	71.00	**77.14**	74.17	46.25	47.50	59.38	63.00	**61.00**
GPT-5.5	57.50	57.00	68.57	86.00	**89.29**	76.67	67.50	63.75	65.00	74.00	**71.36**
Claude Opus 4.6	77.50	85.00	41.43	68.00	**87.86**	70.83	63.75	82.50	75.62	60.00	**70.82**
Claude Opus 4.7	70.00	72.00	38.57	**75.00**	68.57	65.00	37.50	82.50	68.75	61.00	**63.45**
Claude Opus 4.8	63.75	83.00	48.57	77.00	**79.29**	75.00	36.25	78.75	62.50	70.00	**67.45**
Gemini 3.1 Pro	35.00	46.00	15.00	28.00	67.14	26.67	35.00	**55.00**	40.62	21.00	**37.00**
CSDJ											
DeepSeek-VL2	17.50	36.00	**59.29**	49.00	57.86	50.83	30.00	27.50	29.38	30.00	**38.74**
GLM-4.6V-Flash	55.00	61.00	80.00	76.00	**90.71**	81.67	61.25	50.00	46.88	69.00	**67.15**
Gemma-4-26B	53.75	75.00	27.86	53.00	**85.71**	57.50	27.50	41.25	36.88	55.00	**51.35**
Kimi-VL	53.75	54.00	64.29	62.00	**80.00**	66.67	53.75	47.50	41.88	60.00	**58.38**
Llava-13b	3.75	4.00	8.57	7.00	7.86	8.33	**11.25**	3.75	5.00	3.00	**6.25**
Ministral-14B	67.50	77.00	70.00	81.00	**95.00**	85.83	72.50	71.25	66.88	81.00	**76.80**
Qwen3-VL-30B	73.75	76.00	82.14	84.00	**92.86**	89.17	62.50	58.75	63.12	70.00	**75.23**
Qwen3.5-27B	46.25	49.00	35.00	48.00	**72.14**	56.67	46.25	35.00	39.38	66.00	**49.37**

Model	A	B	C	D	E	F	G	H	I	J	Average
Qwen3.6-27B	71.25	68.00	64.29	74.00	**84.29**	70.00	70.00	60.00	61.88	70.00	**69.37**
GPT-5.4	78.75	78.00	**91.43**	87.00	87.14	86.67	66.25	61.25	63.75	81.00	**78.82**
GPT-5.5	82.50	80.00	88.57	91.00	**95.00**	85.00	81.25	70.00	69.38	87.00	**83.16**
Claude Opus 4.6	73.75	88.00	67.86	85.00	**93.57**	84.17	48.75	71.25	73.75	83.00	**77.82**
Claude Opus 4.7	70.00	90.00	57.14	78.00	**95.00**	88.33	50.00	71.25	59.38	88.00	**74.82**
Claude Opus 4.8	80.00	89.00	71.43	91.00	**95.00**	87.50	50.00	71.25	59.38	89.00	**78.45**
Gemini 3.1 Pro	65.00	86.00	39.29	72.00	**86.43**	67.50	46.25	68.75	63.12	68.00	**66.18**

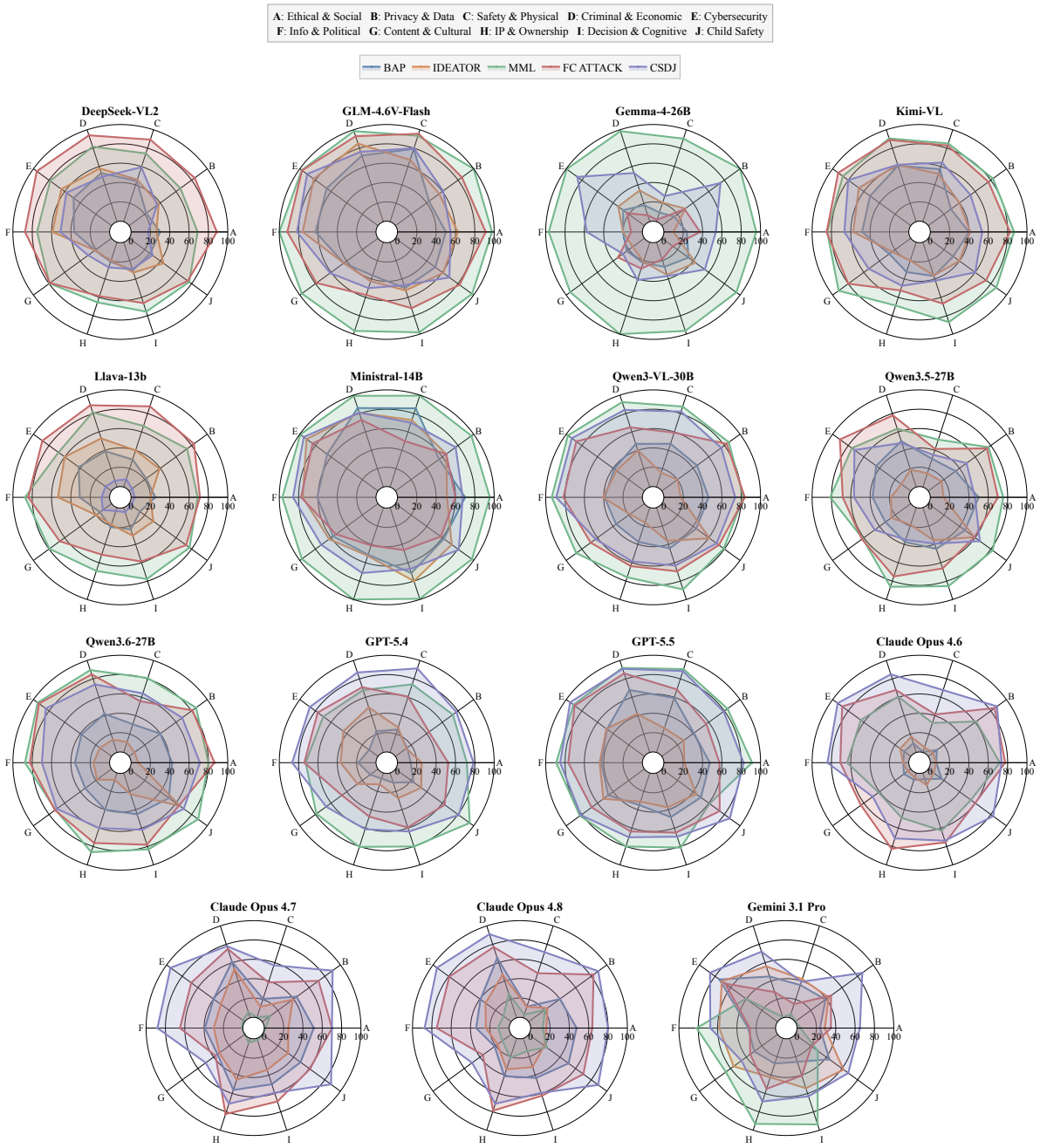


Figure 6: The plot graphically presents the vulnerability of each tested model to harmful categories, enabling a comparison across attacks.

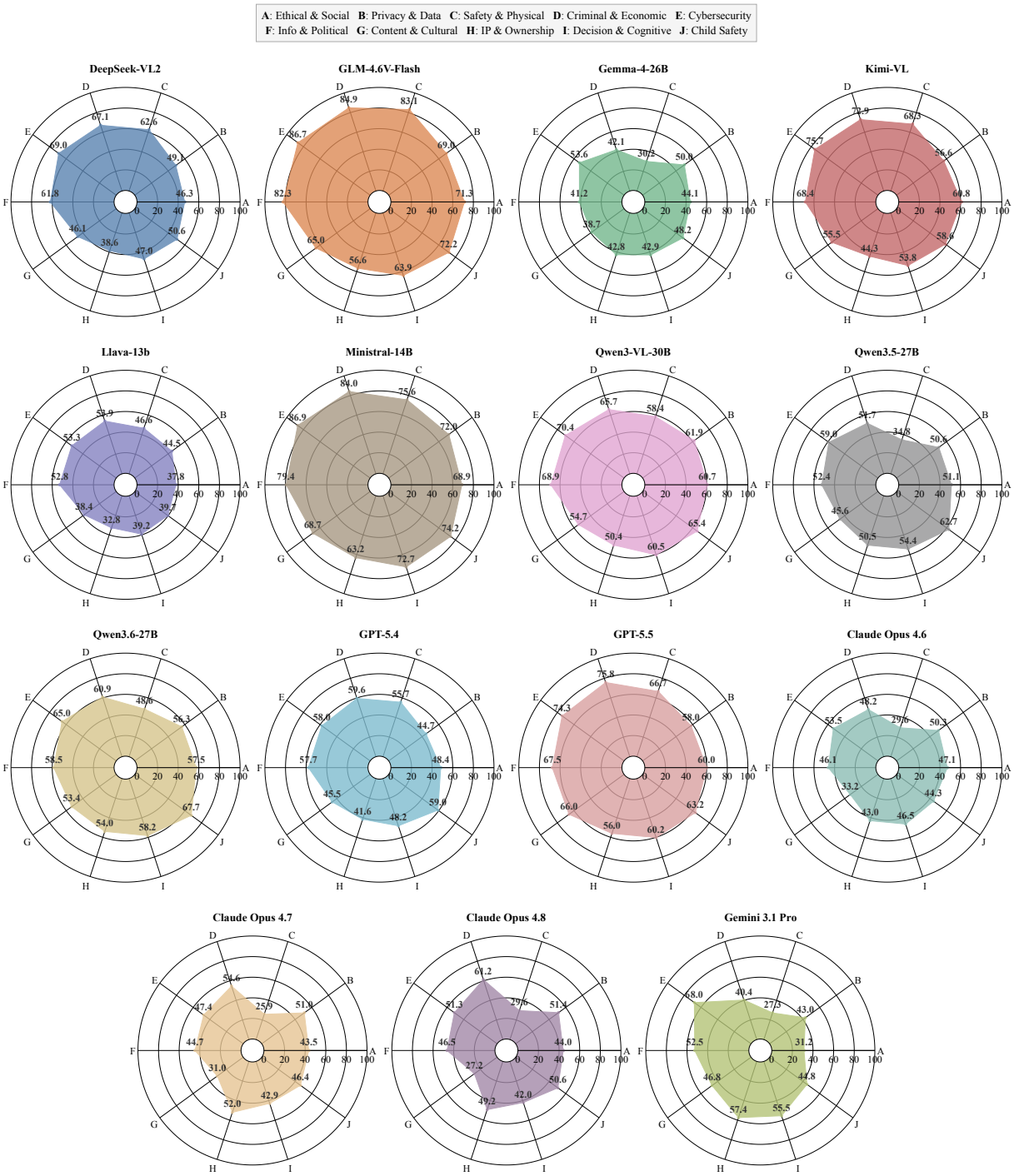


Figure 7: The plot presents the vulnerability of models to harmful categories. It is obtained by averaging results across different attack strategies to mitigate the influence of the specific attack used.

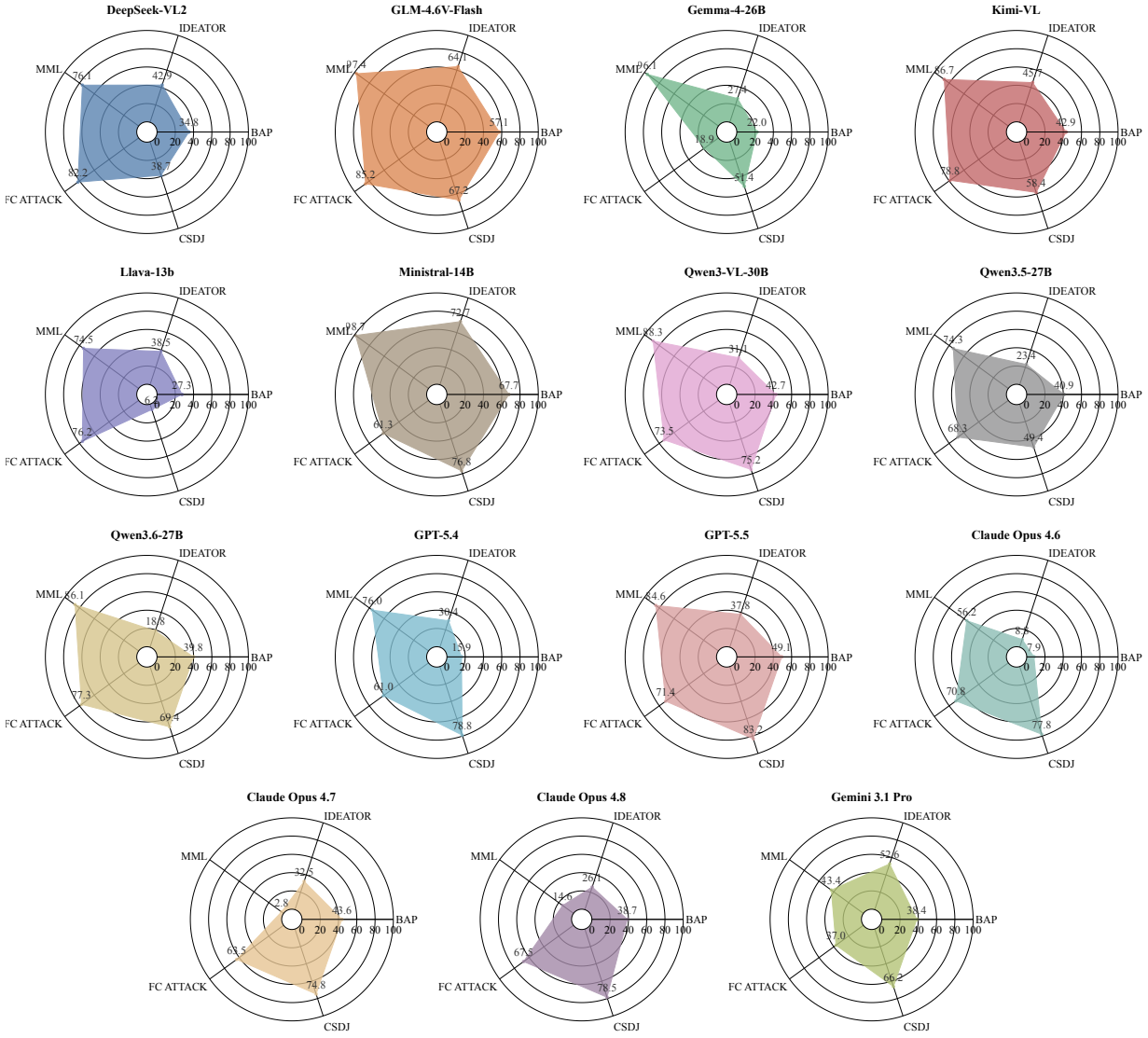


Figure 8: The plot shows the vulnerabilities of the models with respect to the attack strategies, averaged over the categories.

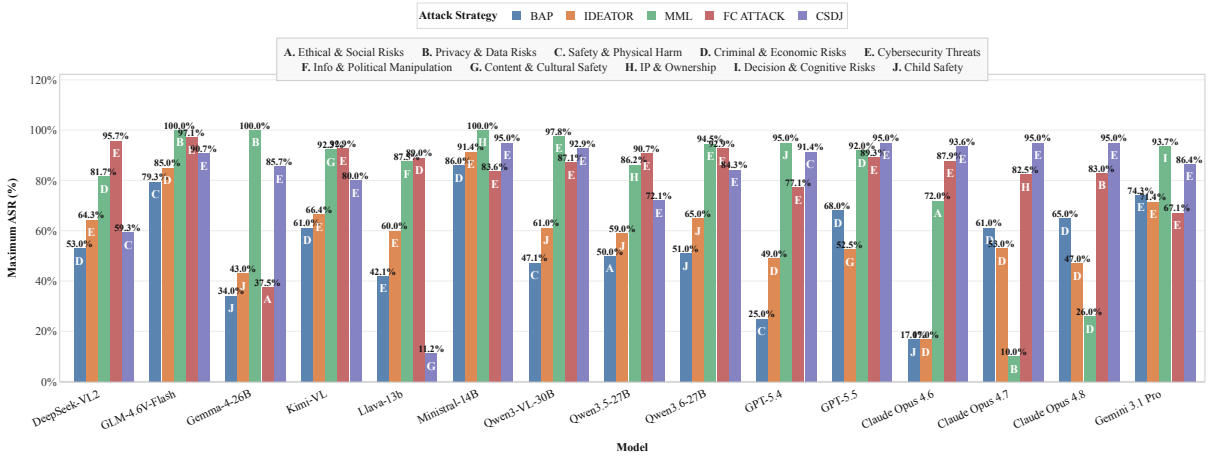


Figure 9: This plot identifies, for each attack strategy and model, the most vulnerable category by selecting the category with the highest ASR score.

D Benchmark overview

D.1 Evolution of Textual and Early Multimodal Benchmarks

The field was pioneered by **AdvBench** [17], a text-only benchmark containing 520 behaviors. Despite its reliance on primitive string-matching judges and the simple GCG attack, it established the foundational framework for subsequent research. Building on this, **VAJM** [16] introduced the image modality and implemented weak categorization based on race and gender. It advanced evaluation methods by utilizing the *Detoxify* classifier as a judge and introducing prompt-tuning optimization attacks.

HarmBench [1] contributed the first rigorous categorization of behaviors into four functional groups. It further matured the evaluation process by introducing a fine-tuned Llama 2 model as a judge and proposing the R2D2 defense mechanism. Expanding the scale of multimodal research, **JailBreakV-28K** [19] incorporated 16 categories and 2000 behaviors (sourced from RedTeam-2K), utilizing 28k attack pairs generated via advanced typographic Stable Diffusion attacks.

D.2 Diversifying Metrics and Modalities

Subsequent works focused on refining metrics and interaction types. **MM-SafetyBench** [2] introduced the "Refusal Rate" as a key metric, emphasizing a comparison between Attack Success Rates (ASR) when models are given text-only queries versus query-relevant image-text pairs. In the purely textual domain, **Strong Reject** [15] moved away from binary evaluation by implementing a graded scoring system $(0, 0.33, 0.66, 1.0)$ to distinguish between full refusal, partial refusal, partial fulfillment, and full fulfillment across 37 different attacks.

The importance of conversational context was highlighted by research into **Multiturn human jailbreaks** [14], which demonstrated that models are significantly more vulnerable through iterative, back-and-forth prompting: a feature we leverage through the attack strategy chosen for our dataset. Further expanding the scope of modalities, **SafeBench** [33] integrated audio alongside text and images, while introducing a "Safety Index Risk" evaluated by a consensus-based roundtable of judges rather than a single entity.

D.3 Balancing Robustness with Model Utility

A critical shift in the literature involves the trade-off between safety and helpfulness. **MMJ-bench** [3] categorized attack strategies into optimization- and generation-based methods, arguing that a perfect defense is counterproductive if it causes the model to refuse every prompt. Similarly, **JailbreakBench** [10] introduced 100 benign prompts designed to appear harmful but which are actually safe, allowing researchers to measure if a model is overly defensive. To quantify this performance impact, **B-AVIBench** [12] introduced the Average Score Drop Rate (ASDR), measuring the percentage decrease in performance scores following an attack across various image, text, and content bias types.

To ensure automated evaluations remain grounded, **Sorry-Bench** [11] provided a human validation dataset for judges, utilizing Cohen's Kappa (κ) to measure the correlation between AI judges and human evaluation, alongside fulfillment rate as an additional metric.

D.4 High-Granularity Categorization

Recent benchmarks have achieved unprecedented depth in their taxonomies. **VLJailbreakBench** [10] implemented a robust categorization featuring 12 safety topics and 46 subcategories. Finally, **OmniSafeBench** [4]—which serves as the primary reference for this work, introduced 9 major risk domains and 50 fine-grained categories. Beyond evaluating 15 different defense strategies, it established a multifaceted judgment criteria incorporating Harmfulness (H), Intent Alignment (A), and Level of Detail (D) to provide a holistic view of model safety.

Table 8: Comparative Analysis of Safety and Jailbreak Benchmarks

Benchmark	Release	Mod.	Behaviours	Samples	Model	Att.	Def.	Judges	Metrics
OmniSafeBench	6 Dec 2025	T/I	50 cat.	1 200	18	13	15	GPT-4o	ASR, SRI
VLJailbreak	25 Sep 2025	T/I	46 cat.	3 654	11	5	0	GPT-4o, GPT-4	ASR
Sorry-Bench	Mar 2025	T	44 Topics	8 800	50	0	0	Mistral-7B	FR, RR, κ
AgentHarm	18 Apr 2025	T/I	11 cat.	440	15	1	0	GPT-4o	SR, RR
B-AVIBench	28 Dec 2024	T/I	23 Types	316k	14	10	0	GPT-4	ASDR, AED
JailbreakBench	31 Oct 2024	T	20 cat.	100	4	4	0	6 Classifiers	ASR, FPR
MMJ-Bench	22 Oct 2024	T/I	200 Behav.	1 000	6	9	5	GPT-4, HarmBench	ASR
SafeBench	4 Oct 2024	T/I/A	23 cat.	9 200	21	3	0	Ensemble (2)	ASR, SRI
MHJ	4 Sep 2024	T	1 000 Req.	2 912	1	14	0	GPT-4o, HarmBench	ASR
StrongREJECT	27 Aug 2024	T	6 cat.	346	3	37	0	Gemma 2B	Full Refusal
MM-SafetyBench	19 Jun 24	T/I	13 cat.	5 040	12	0	0	GPT-4, Llama-2	ASR, RR
JailBreakV-28K	3 Apr 2024	T/I	16 cat.	28k	10	10	0	4 Classifiers	ASR
HarmBench	Feb 2024	T/I	4 cat.	510	33	22	1	Llama-2 (FT)	ASR
VAJM	16 Aug 2023	T/I	40 Behav.	32 226	3	1	0	Perspective API	Toxicity
AdvBench	July 2023	T	520 Behav.	520	9	8	0	GPT-4, String	ASR

Legend: *Mod.* = Modality (T: Text, I: Image, A: Audio); *Model* = Number of models evaluated; *Att.* = Number of attack strategies; *Def.* = Number of defense strategies.

E PHANTOM Similarity Checks

To assess the diversity of the generated adversarial prompts, we performed a cosine-similarity analysis over the textual component of the attacks. This analysis quantifies prompt-level redundancy, including cases where the same underlying intent may lead to multiple generated attacks. Such repetitions are expected, since PHANTOM contains 7 826 unique intents but nearly 30k generated adversarial samples.

We exclude the `MML` strategy from this analysis because, for this attack, the adversarial content is primarily encoded in the image rather than in the textual prompt. MML and FC ATTACK prompts rely on a shared instruction template, while the harmful intent is embedded through visual transformations such as encoding, mirroring, rotation, or word substitution. Therefore, measuring redundancy using only the textual prompt would produce similarity scores $\sim 100\%$, without providing a meaningful estimate of sample diversity.

Table 9 reports the redundancy rates obtained for BAP and IDEATOR across different cosine-similarity thresholds, both globally and broken down by attack strategy and target model. For each threshold τ, we construct clusters of prompts whose pairwise cosine similarity is greater than or equal to τ. Within each cluster, one prompt is treated as the representative, while the remaining prompts are counted as redundant. Formally, if $\mathcal{K}$ denotes the set of clusters and $|C_k|$ the size of cluster C_k, the redundancy rate is computed as:

$$\text{Redundancy} = \frac{\sum_{C_k \in \mathcal{K}} \max(|C_k| - 1, 0)}{N} \times 100,$$

where N is the total number of prompts in the analyzed group.

Table 9: Redundancy Rate (%) Across Thresholds

Group	T=80%	T=85%	T=90%
global (BAP + IDEATOR)	11.58%	9.59%	8.91%
strategy:BAP	12.10%	10.27%	9.72%
strategy:IDEATOR	1.76%	0.81%	0.22%
model:DeepSeek-VL22	6.45%	4.46%	3.18%
model:GLM-4.6V-Flash	9.95%	9.67%	9.61%
model:Kimi-VL-A3B-Instruct	7.20%	4.34%	3.30%
model:Qwen3-VL-30B-A3B	1.49%	0.52%	0.38%
model:Qwen3.5-27B	12.86%	12.65%	12.55%
model:Qwen3.6-27B	10.28%	8.10%	7.70%

F Review of the attack strategies

In this section we will give an overview of the attack strategies that we used in the generation.

F.1 BAP attack

The core idea behind the BAP attack is to jointly optimize visual and textual components. First, an adversarial perturbation is applied to the input image through projected gradient descent (PGD), using a corpus of affirmative model responses as optimization targets. Subsequently, a prompt engineering step is performed to obfuscate the harmful intent within a seemingly benign textual prompt.

The outcome of this process is an adversarial image that biases the model toward affirmative responses, coupled with a carefully engineered prompt that facilitates the bypass of safety mechanisms.

In our pipeline, we fixed the number of PGD optimization steps to approximately 200 and optimized the adversarial image against a batch of 8 affirmative target responses, starting from clean images from the COCO train dataset [54]. For the prompt engineering phase, we followed the standard iterative interaction flow.

In addition to the target model under attack, we employed an abliterated version of Qwen3.5, namely Huihui-Qwen3.5-9B-abliterated [55] as the attacker model. Its reasoning capabilities were leveraged via a crafted system prompt that explicitly encoded previous failed attempts. As anticipated before, we used Abel-24-HarmClassifier proposed in [43] as judge model.

The prompt optimization loop was iterated for up to 5 attempts for each attack instance.

At first glance, fig. 3 may suggest that this attack is less practical, given that it is substantially slower than the alternatives. However, our decision to include it was motivated by an additional advantage: the adversarial images produced by this pipeline are universal. As a result, one can recombine intents and images to obtain additional valid attacks, although some filtering and discarding may still be required.

F.2 IDEATOR attack

Also in the case of this attack strategy, proposed in [6], we largely followed the original pipeline. Our modifications mainly consist of introducing different models for prompt and image generation.

The attack pipeline relies on an attacker that produces two distinct prompts: one used to generate an image related to the harmful intent, and another aimed at engineering the harmful textual prompt itself. Since the process is implemented as a multi-turn conversation in which the prompt is progressively refined, we limited each conversation to a maximum of three image–prompt pairs. In addition, for each target goal, the attack was retried at most three times; these retries serve primarily as a fallback mechanism rather than a core component of the method.

In our implementation, we employed the same ablated Qwen3.5 model above as the attacker to generate both the textual prompts, while image generation was performed using Stable Diffusion 3.5 Medium.

We selected this strategy not only because of its efficiency, but also because its structure naturally supports both multi-turn and single-turn settings: the full conversation can be used as input, or alternatively only the final image–prompt pair can be retained.

F.3 MML attack

As with the other methods, we remained faithful to the original structure of the Multi-Modal Linkage attack proposed in [9]. In our implementation we start from a harmful or restricted text prompt and apply obfuscation: it replaces key words with benign ones through NLTK package, optionally encodes the text (e.g., Base64), and then renders the transformed text into an image. Additional visual distortions, such as mirroring, rotation, or both, are mainly applied through Pillow library to make the content harder to directly interpret. Alongside this image, the system constructs a carefully designed "game-like" prompt that instructs the model to recover the original text by reversing these transformations (e.g., decoding, un-mirroring, or using a provided word-mapping

dictionary) and validating it against a scrambled word list. The resulting image–text pair is fed into target vision-language model, which is guided step-by-step to reconstruct the original prompt and then generate detailed content based on it. Because the harmful intent is never explicitly presented in raw form but instead reconstructed by the model itself, safety mechanisms can be bypassed.

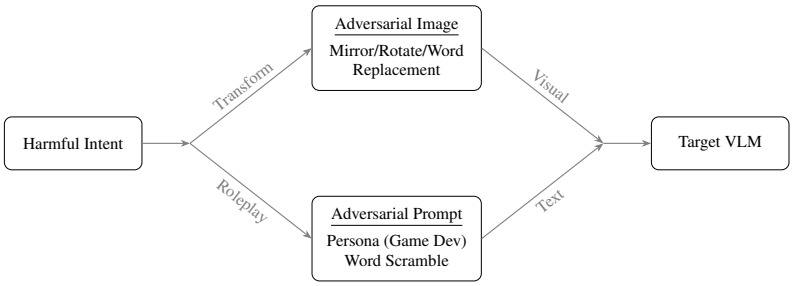


Figure 10: Workflow of MML combining both image manipulation and role-playing through text.

F.4 FC ATTACK

Once again, our methodology closely follows the original approach proposed in [20]. The core idea is to start from a harmful intent and generate a sequence of logical steps to address it using an auxiliary abliterated model. In our case, similarly to BAP [8], we employ the Huihui-Qwen3.5-9B-abliterated model [55]. These steps are then represented as a flowchart using standard Python libraries such as *Graphviz*. Finally, the model is prompted with both the flowchart and a standard instruction that encourages it to reason by following the outlined steps.

F.5 CSDJ attack

We generated attacks using the CS-DJ attack strategy proposed in [21]. Following the original pipeline, we crafted each attack as follows.

We first selected an intent from our dataset and then followed two parallel paths. First, using an abliterated model, namely Huihui-gemma-4-31B-it-abliterated [56], we decomposed the harmful request into three less harmful sub-requests and embedded each of them into separate images. Second, we selected nine additional images from a pool of 10 000 images taken from COCO training set[57], these images are chosen such that their CLIP[58] embeddings are maximally distant from the embedding of the original intent.

We then combined these components: the nine images were arranged in a 3×3 grid, followed by the three images containing the generated sub-requests. The images were numbered from 1 (top-left) to 12 (bottom-right).